

Chapter 02

Critical Infrastructure

Operational Technologies (OT)

Controls physical operations

Used in industrial environments

Failure creates safety risks

Supports critical systems

Industrial Control Systems (ICS)

Central to operational technology

Manage system operations

Used in infrastructure systems

Control industrial processes

ICS Safety Controls

Prevent unsafe outputs

Enforce system rules

- Traffic lights coordination

Designed for safe operation

ICS Automation and Data Flow

Automate system operations

Control machinery processes

Communicate status updates

Integrate with SCADA systems

SCADA Systems Overview

Supervisory Control and
Data Acquisition

Monitor system activity

Provide operational visibility

Track system status

SCADA Security Importance

Control critical infrastructure

Must limit external access

Require strong protections

Prevent system compromise

Stuxnet Attack Example

Targeted ICS and SCADA systems

Affected nuclear infrastructure

Spread despite isolation

Demonstrates real-world threat

Critical Infrastructure Management

Systems must remain operational

Require strong protections

Support essential services

Demand continuous monitoring

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Types of Systems

Virtualization

Multiple VMs on one machine,
managed by a hypervisor

- Type 1: hypervisor IS the OS — minimal overhead (e.g., Hyper-V)
- Type 2: runs atop a host OS — more overhead (e.g., VMware)

IaC and Containerization

Virtualized network devices
managed through code, paired
with SOAR

Containerized apps run in
isolation — easily relaunched if
they crash

Cloud Service Models

SaaS: provider manages everything — you just use the software

PaaS: provider gives you a platform to build on

IaaS: hardware only — think of it as Hardware as a Service

Hybrid Cloud and MDM

Hybrid cloud blends cloud flexibility with on-premises control

MDM software monitors, secures, and manages mobile device fleets

- Enforces PINs, restricts cameras, limits unapproved apps

EDR and Zero Trust

EDR monitors endpoints and protects devices on or off the network

Zero trust: never trust, always verify — temporary tickets required

- No ticket, no connection — like a hall pass in school

SASE

SASE combines SD-WAN, zero trust, and security tools in one cloud suite

Security Service Edge handles access, monitoring, and protection